

Automate backups with Amazon Data Lifecycle Manager

To Begin with the Lab

Summary of the Lab

This lab walks through creating an EC2 instance with a Linux AMI on AWS. After launching the instance and assigning tags, you use **Elastic Block Store (EBS) Lifecycle Manager** to automate snapshot and AMI backup policies. Three types of policies are demonstrated:

1. **EBS Volume Snapshot Policy** – Captures data from an attached volume; useful for restoring volume data but not the whole instance.
 2. **EBS Snapshot from Volume Policy** – Takes volume-level snapshots, restorable on other instances.
 3. **EBS-backed AMI Policy** – Creates full bootable AMIs of EC2 instances for full backup, cloning, or disaster recovery.
- Let's create an instance.
 - Open the Amazon EC2 console at [AMAZON EC2](#).
 - Choose Launch instance.
 - (Optional) Under Name and tags, for Name, enter a name to identify your instance.
 - Under Application and OS Images (Amazon Machine Image), choose a Linux operating system, and then for Amazon Machine Image (AMI), select a Linux AMI.

The screenshot displays the Amazon EC2 console's 'Launch instance' wizard. The 'Name and tags' section at the top has a text input field for 'Name' containing 'linuxvm'. Below this, the 'Application and OS Images (Amazon Machine Image)' section shows a grid of operating system logos. 'Ubuntu' is highlighted with a red box, and the 'Quick Start' tab is also highlighted. The 'Summary' panel on the right provides a overview of the configuration: 1 instance, Canonical Ubuntu 24.04 LTS (HVM) SSD Volume Type (ami-020c3a7c55df1f615) as the software image, t2.micro as the virtual server type, a new security group for the firewall, and 1 volume(s) of 8 GiB for storage. A 'Free tier' notice indicates that the user is eligible for 750 hours of free t2.micro instance usage per month. At the bottom right, there are 'Cancel', 'Launch instance', and 'Preview code' buttons.

- Under Instance type, for Instance type, select an instance type or keep the default.

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t2.micro Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- Under Key pair (login), for Key pair name, choose an existing key pair or create a new one.
- Click on the create new key pair.
- Choose the private key file format as '.ppk'.
- Click create key pair.

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

A key pair to securely connect to your instance. Ensure that you have the private key file on your local computer.

Key pair name [Info](#)

Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☐ .pem
For use with OpenSSH

☒ **ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

[Cancel](#) [Create key pair](#)

- Here I am not making any other changes Networking setting and Configure storage.
- Launch instance

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File Systems [Edit](#)

► **Advanced details** [Info](#)

tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)

- Now select the instance
- Go to Tags.

EC2

Dashboard
EC2 Global View
Events

▼ **Instances**
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations

Instances (1/2) [Info](#)

Last updated 3 minutes ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	linuxvm	i-023b05e63c0afd55e	Running	t2.micro	Initializing	View alarms	us-east-1b	ec2-34-2f...
☐	ub1	i-0d1837fbbd1a1645b	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	ec2-3-82...

i-023b05e63c0afd55e (linuxvm)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

- Click on manage tags.

i-023b05e63c0afd55e (linuxvm) ⚙️ ▼

Details | Status and alarms | Monitoring | Security | Networking | Storage | **Tags**

Tags Manage tags

Q

Key | Value

Name	Value
linuxvm	

- Add a new tag.
- Click save.

Manage tags info

A tag is a custom label that you assign to an AWS resource. You can use tags to help organize and identify your instances.

Key Value - optional

Q Name X Q linuxvm X Remove

Q dlm X Q true X Remove

Add new tag

You can add up to 40 more tags.

Cancel Save

Tags Manage tags

Q

Key | Value

Name	Value
linuxvm	
dlm	true

- Now go to Lifecycle Manager inside Elastic Block Storage.
- Here we are making policy for Snapshot creation using EBS snapshot policy.
- Choose custom policy.
- Click on the next step

Savings Plans | Reserved Instances | Dedicated Hosts | Capacity Reservations | **Images** | AMIs | AMI Catalog | **Elastic Block Store** | Volumes | Snapshots | **Lifecycle Manager**

Amazon Data Lifecycle Manager
Automate the creation, retention, copy and deletion of snapshots and AMIs

Amazon Data Lifecycle Manager provides a simple, automated way to back up data stored on Amazon EBS volumes.

Create new lifecycle policy

Create custom or default policy

☒ Custom policy
☐ Default policy

Policy type

EBS snapshot policy

Next step

- **When we Use EBS Snapshot Policy on a Volume Attached to an EC2 Instance:**
 - It creates a backup of that volume's current state, even if it's the root volume.
 - You can restore the volume data, but you cannot recreate the full EC2 instance from this snapshot alone.
 - If it's a root volume, you'll need to manually recreate a bootable image (complex and error-prone).
- In the specify settings
- Choose instance
- Choose the key and value and click add.
- It will add below as shown.

Target resources [Info](#)
Specify the resources that are to be targeted by this policy.

Target resource types
Select the type of resources that are to be targeted.

☐ Volume

☒ Instance

Target resource tags
All resources of the selected type that have at least one of these tags will be targeted by the policy.

44 tags remaining of 45.

- Give the description.

Description

Policy description

- Leave all other steps on default
- click on the next.
- In configure schedule, you can set the schedule according to you need.
- Also, you can add another schedule if you want.

Schedule details [Info](#)

You can add 3 more schedules to this policy.

Schedule name

Frequency

Every

Starting at

UTC

Retention type

Keep

- And then click review policy.
- Then click create policy.
- Lifecycle policy is successfully created.

☒ Successfully created lifecycle policy. [policy-0e1602d46df11487](#)

Data Lifecycle Manager (1) [Info](#)

☐	Name	Policy ID	Description	Policy type	Policy state
☐		policy-0e1602d46df11487	my-linux-vm-backup	EBS snapshot policy	☒ Enabled

- If we want to create the policy through the volume
- Same steps are there, go to volume

- Select the volume
- Click on the tags
- Click on the manage tags.

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone
☐		vol-0e4171c9b3dd604d7	gp3	2 GiB	3000	125	snap-02bc826...	2025/07/01 07:24 GMT+5:...	us-east-1d
☐		vol-0d26fb1a55510ee11	gp2	8 GiB	100	-	snap-0040175...	2025/07/01 07:06 GMT+5:...	us-east-1b
☒		vol-0211a748515fe2362	gp3	8 GiB	3000	125	snap-0bc1d35...	2025/07/01 12:21 GMT+5:...	us-east-1b
☐		vol-0fad42d2ce9f990d2	gp3	2 GiB	3000	125	-	2025/07/01 07:06 GMT+5:...	us-east-1b

Volume ID: vol-0211a748515fe2362

Details
Status checks
Monitoring
Tags

Tags

Key
Value

Manage tags

- Click on add tag
- Enter the Key and value
- Click save.

Manage tags
Info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional
Remove

Add tag

You can add 49 more tags.

Cancel
Save

Now go to Lifecycle Manager inside Elastic Block Storage.

- **When we Use EBS Snapshot Policy on an EBS Volume:**
 - A snapshot of the selected EBS volume is created.
 - It captures all the data on that volume at the time of the snapshot.
 - You can later create a new volume from the snapshot and attach it to any instance.
 - It does not include instance-level settings or make the volume bootable.
- Here we are making policy for Snapshot creation using EBS snapshot policy.
- Choose custom policy.
- Click on the next

Select policy type [Info](#)

The Data Lifecycle Manager default policy for EBS snapshots backs up all volumes in this Region that do not have recent snapshots. You can exclude volumes using exclusion parameters if needed. [Learn more](#)

Select whether to use a custom or default policy

☒ **Custom policy**
Data Lifecycle Manager custom policies let you target specific volumes or instances based on their assigned tags. You can add up to 4 schedules per policy and configure advanced features, such as fast snapshot restore, snapshot archiving, cross-account copying, and pre and post scripts.

☐ **Default policy - new**
Data Lifecycle Manager default policies back up all volumes and instances in this Region that do not have recent backups. You can exclude volumes and instances by specifying exclusion tags or other exclusion parameters if needed.

Schedule-based policy

☒ **EBS snapshot policy**
Create a policy that automates the creation, retention, and deletion of EBS snapshots.

☐ **EBS-backed AMI policy**
Create a policy that automates the creation, retention, and deletion of EBS-backed AMIs.

Event-based policy

☐ **Cross-account copy event policy**
Create a policy that automates cross-account copy for snapshots that are shared with your AWS account.

[Cancel](#) [Next](#)

- In the specify settings
- Choose instance
- Choose the key and value and click add.
- It will add below as shown.

Target resources [Info](#)

Specify the resources that are to be targeted by this policy.

Target resource types
Select the type of resources that are to be targeted.

☒ **Volume**
☐ Instance

Target resource tags
All resources of the selected type that have at least one of these tags will be targeted by the policy.

[Add](#)

linuxEBS [X](#)

true

44 tags remaining of 45.

- Give the description.

Description

Policy description

volume-backup

- Leave all other steps on default
- click on the next.
- In configure schedule, you can set the schedule according to you need.
- Also, you can add another schedule if you want.

Schedule details [Info](#)

[Remove schedule](#)

[Add another schedule](#)

You can add 3 more schedules to this policy.

Schedule name

Schedule 1

Frequency

Daily

Every

12 hours

Starting at

09:00

UTC

Retention type

Count

Keep

2

snapshots in standard tier

- And then click review policy.
- Then click create policy.
- Lifecycle policy is successfully created.

✔ Successfully created lifecycle policy.
policy-049965ffe952c952c

Data Lifecycle Manager (2) [Info](#)

Last updated
less than a minute ago

[Actions](#)

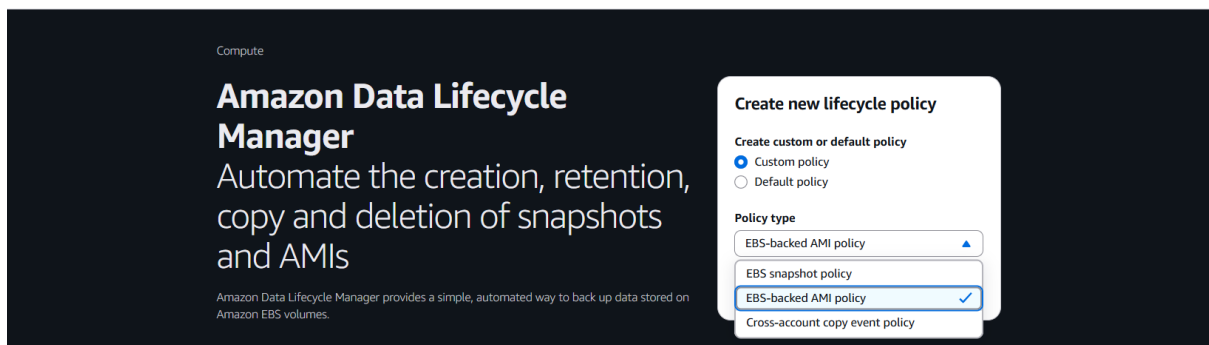
[Create lifecycle policy](#)

All policies

Find policy by attribute or tag

☐	Name	Policy ID	Description	Policy type	Policy state
☐		policy-0e1602d46fdf11487	my-linux-vm-backup	EBS snapshot policy	✔ Enabled
☐		policy-049965ffe952c952c	volume-backup	EBS snapshot policy	✔ Enabled

- **When You Use an EBS-backed AMI Policy:**
 - A complete Amazon Machine Image (AMI) is created from the EC2 instance.
 - This includes the root volume (OS, configs) and optionally other attached volumes.
 - The AMI is bootable and can be used to launch new EC2 instances anytime.
 - It's ideal for full backups, cloning, or disaster recovery.
- Now go to Lifecycle Manager inside Elastic Block Storage.
- Here we are making policy for instance from EBS-backed AMI policy.
- Choose custom policy.
- Click on the next step



- In the specify settings
- Choose instance
- Choose the key and value and click add.
- It will add below as shown.

Specify settings

Target resources [Info](#)

Specify the resources that are to be targeted by this policy.

Target resource tags

All resources of the selected type that have at least one of these tags will be targeted by the policy.

dln true

44 tags remaining of 45.

Give the description.

Description

Policy description

my-linux-machine

- Leave all other steps on default except instance reboot option
- Choose yes in the instance reboot option.

Instance reboot

Indicate whether instances should be rebooted before AMI creation. Choose No to prevent the targeted instances from being rebooted. This could cause data consistency issues. Choose Yes to reboot instances before AMI creation. This ensures data consistency but could result in multiple targeted instances rebooting simultaneously.

Reboot instance at policy run

- ☒ Yes
☐ No

[Cancel](#)

[Next](#)

- click on the next.
- In configure schedule, you can set the schedule according to you need.
- Also, you can add another schedule if you want.

Schedule details [Info](#)

[Remove schedule](#)

[Add another schedule](#)

You can add 3 more schedules to this policy.

Schedule name

Schedule 1

Frequency

Daily

Every

12 hours

Starting at

09:00

UTC

Retention type

Count

Keep

2

snapshots in standard tier

- In the advanced settings
- Check the copy tags from source.

Advanced settings - optional

▼ Tagging [Info](#)

The following tags will be applied to AMIs created by this policy. If you enable cross-Region copy, the tags will not be applied to the copied AMIs. AMIs created by this policy are automatically tagged with the policy ID and schedule name.

☒ Copy tags from source

Variable tags

For variable tags, all resources are tagged with the same tag key, but the tag values vary depending on the source resource.

Select variable tags

instance-id:\${instance-id} X

Additional tags

No tags associated with the resource.

[Add tag](#)

You can add 45 more tags.

- And then click review policy.

► **AMI deprecation** [Info](#)
Enable the deprecation of AMIs created by the schedule.

► **Cross-Region copy** [Info](#)
Enable cross-Region copy to copy AMIs created by this schedule to up to three additional regions.

Cancel Previous **Review policy**

- Then click create policy.
- Lifecycle policy is successfully created